DATA MINING

Outlier Detection in Data Science

- Types
- Challenges
- Method

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What are Outliers?

- Outliers are data points that differ significantly from other observations.
- They can indicate variability, errors, or novel discoveries.
- Outliers affect data distribution, mean, and statistical models.

Outliers can appear because of:

1. Data entry errors → typing mistakes or faulty sensors.
2. Measurement errors → faulty instruments.
3. Natural variation → genuine rare cases (e.g., very tall person, extreme transaction).
4. Sampling issues → mixing data from different populations.

Why Detect Outliers?

- Improve data quality and model accuracy.
- Identify measurement errors or data entry issues.
- Detect fraud, anomalies, or rare events.
- Enhance the reliability of analytics.

Types of Outliers

- 1. Global Outliers: Deviate strongly from the entire dataset.
  Example: A ₹10M transaction in a dataset of ₹10K averages.

- 2. Contextual Outliers: Unusual only in a specific context (e.g., high temperature in winter).

- 3. Collective Outliers: A group of data points collectively behaves abnormally.

Challenges of Outlier Detection

- High Dimensionality – distance metrics lose meaning.
- Dynamic Data – outlier patterns may change over time.
- Noise vs Outlier – hard to distinguish.
- Imbalanced Data – outliers are rare.
- Domain Dependency – what’s abnormal depends on the field.

Overview of Detection Methods

- 1. Statistical Methods – assume data follows a distribution (Z-score, IQR).
- 2. Distance-Based – identify points far from others (k-NN).
- 3. Density-Based – outliers occur in low-density regions (LOF).
- 4. Clustering-Based – points not belonging to any cluster (DBSCAN).

Advanced Outlier Detection Methods

- Isolation Forest – randomly isolates points to find anomalies.
- Autoencoders – reconstruct normal data; errors reveal outliers.
- One-Class SVM – learns boundary of normal data.
- Ensemble Methods – combine multiple algorithms for better accuracy.

Summary and Conclusion

- Outliers reveal hidden patterns or data issues.
- Detection is vital for reliable ML and analytics.
- Choose method based on data type and domain.
- Continuous monitoring ensures data integrity.

Thank you